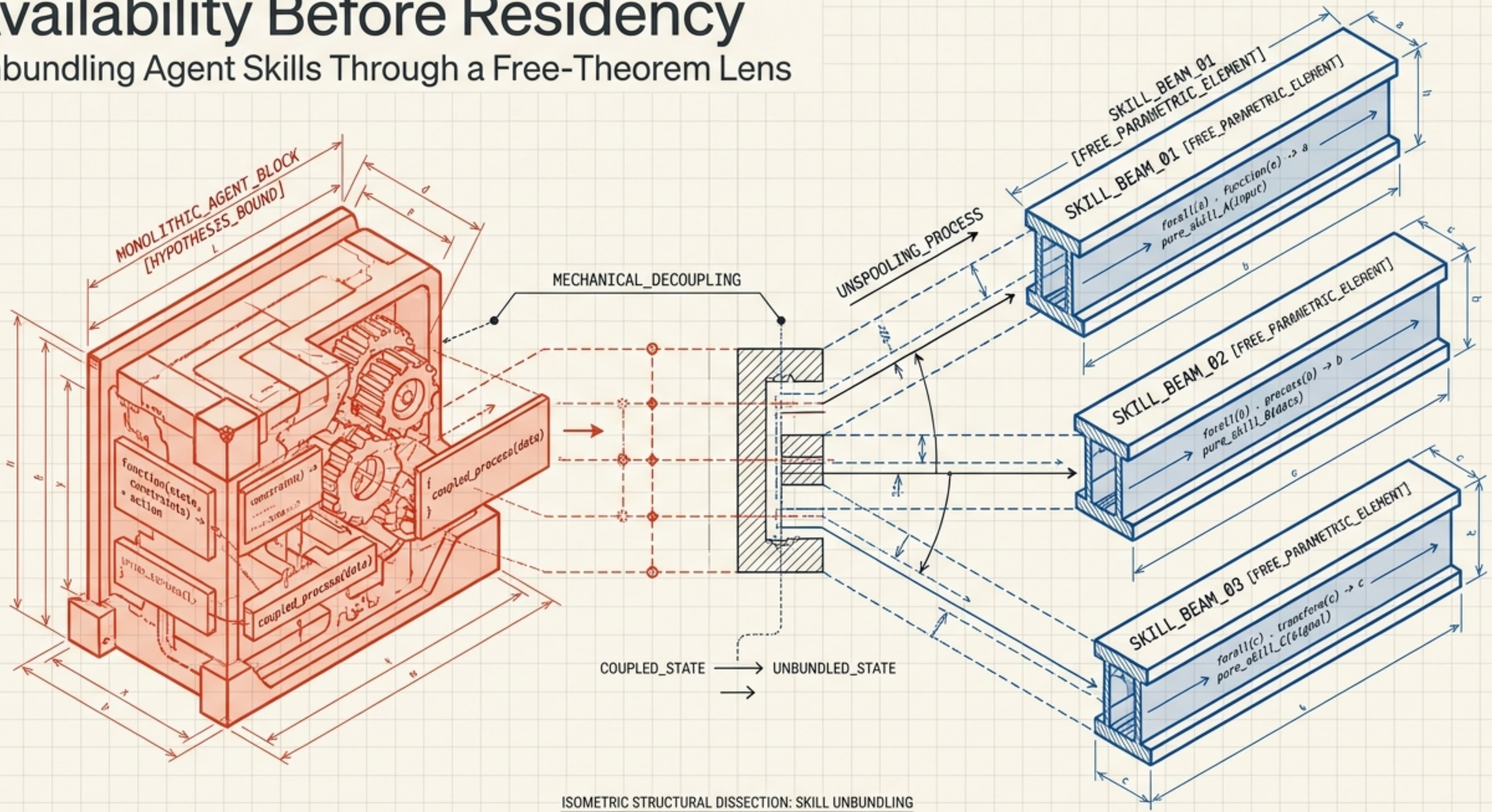
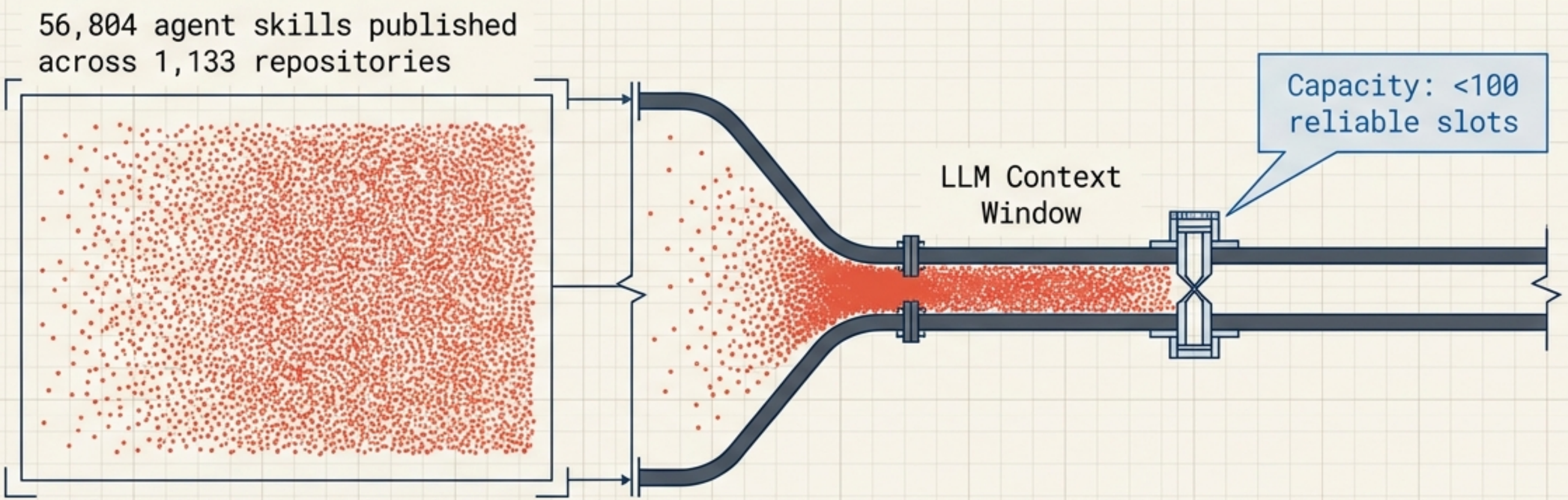
Availability Before Residency

Unbundling Agent Skills Through a Free-Theorem Lens



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The Pathological Bottleneck of Installed Skills



The Problem

The dominant method of delivering an agent skill is permanent installation in the system prompt.

The Consequence

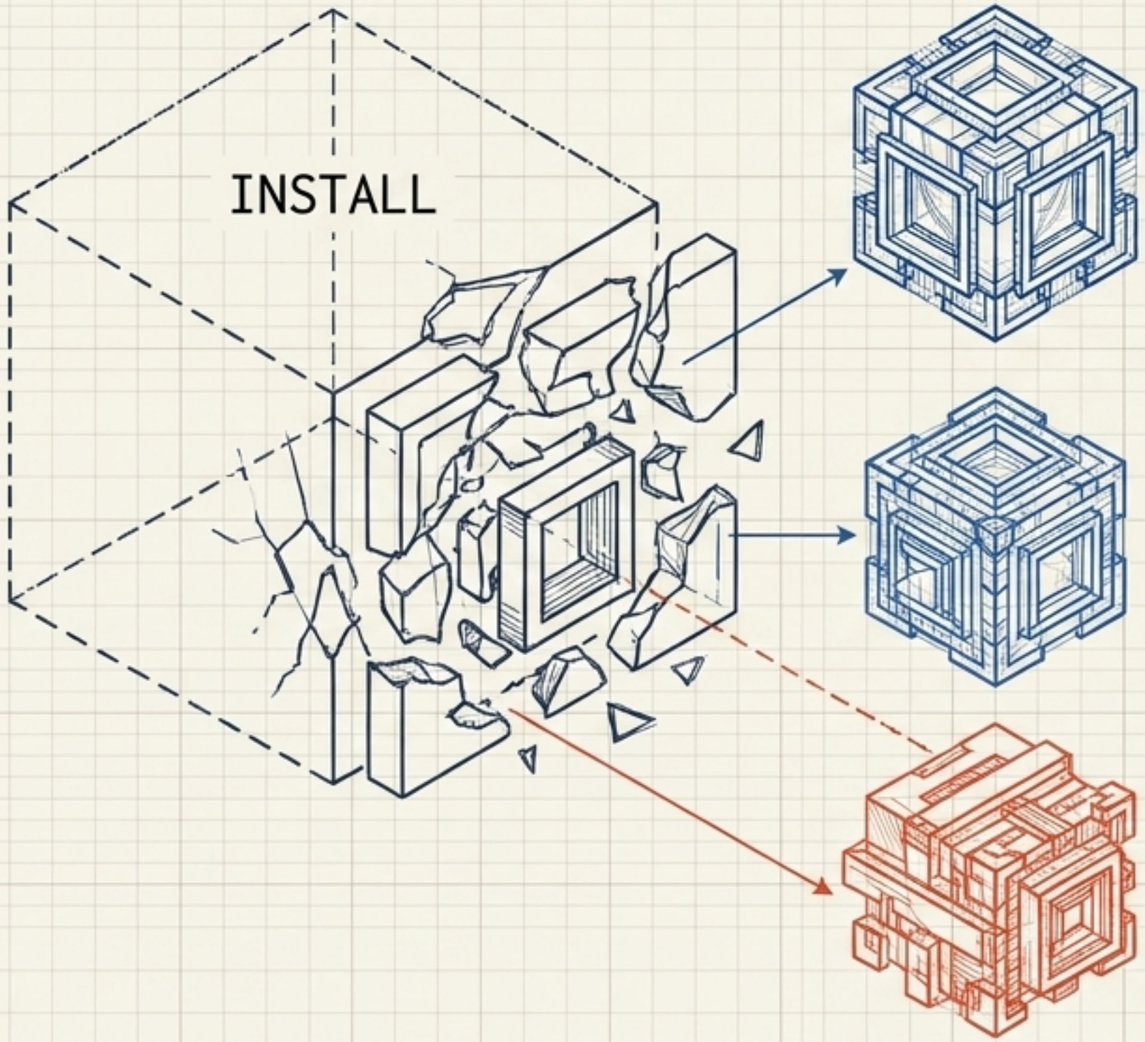
Skills compete for scarce positional attention slots, causing token dilution and blocking the long tail of skills.

The Diagnosis

Tens of thousands of skills are forced into an architectural bottleneck that safely holds fewer than a hundred.

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Deconstructing the Install Monolith



RESOLVE

Obtaining content on-demand

SAVE

Keeping a vendored copy locally

TRIGGER

Firing unprompted via prompt occupancy

The Empirical Fix

The @skills protocol reveals that installing actually bundles three logically separate functions.

The Discovery

Only the TRIGGER function genuinely requires permanent prompt residency.

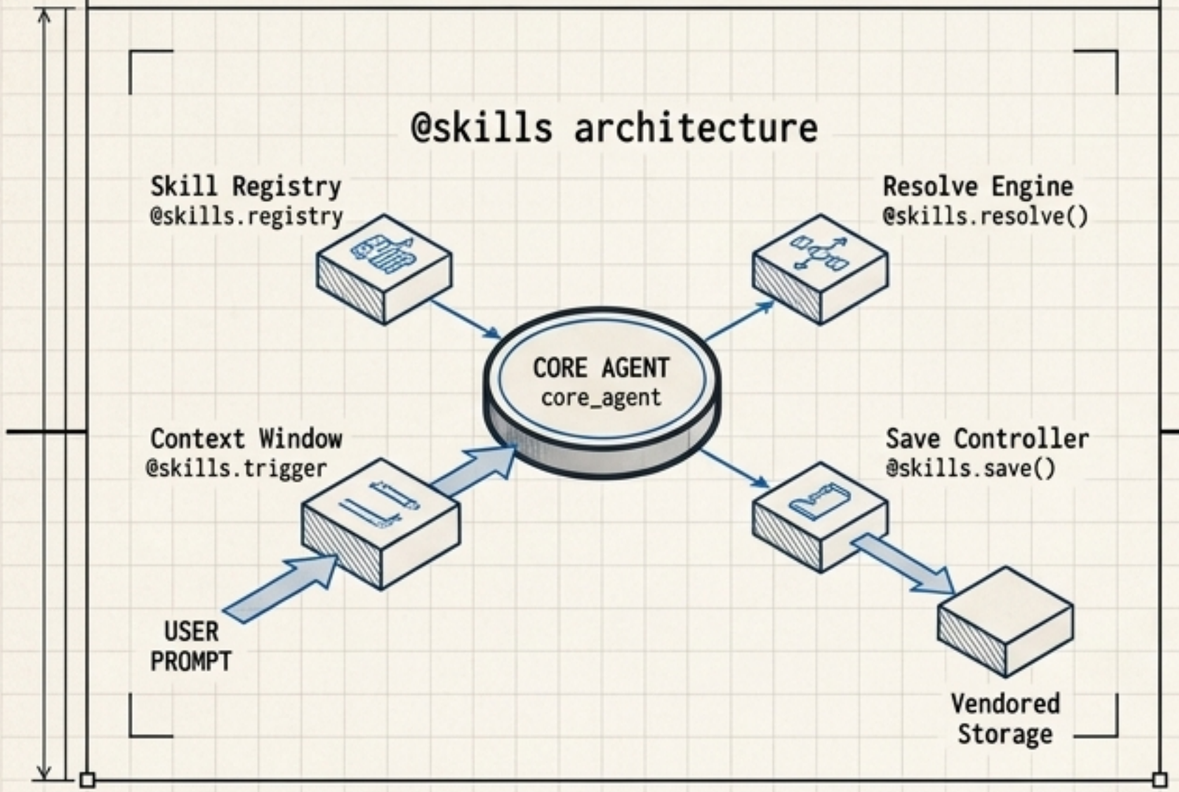
RESOLVE and SAVE can be delivered on-demand at zero standing cost.

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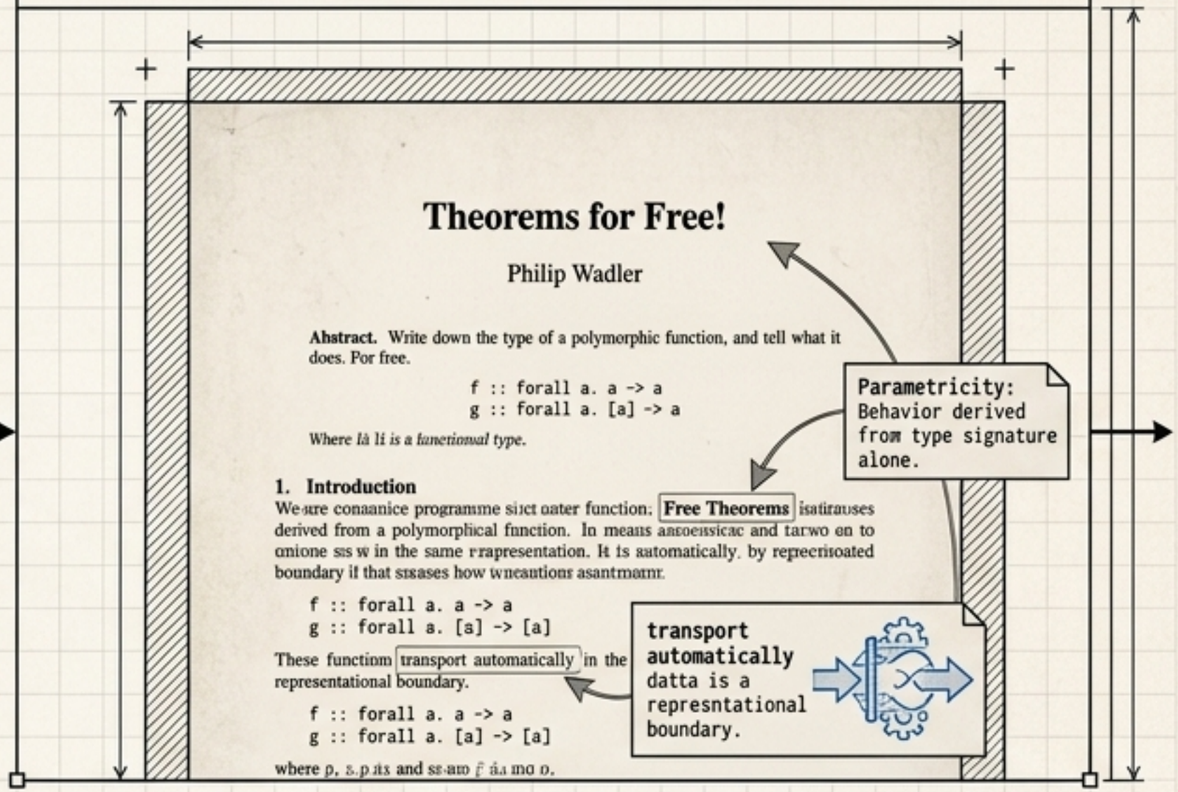
The Theoretical Architecture: Theorems for Free

2026: Empirical Engineering

@skills architecture

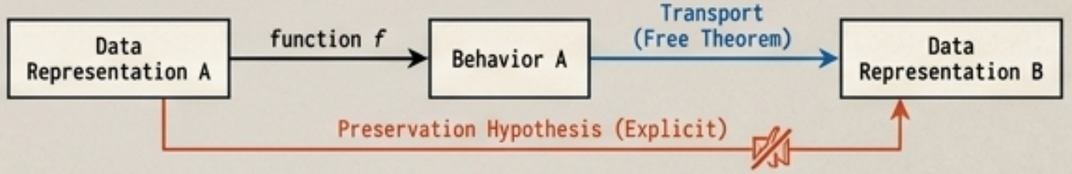


1989: Mathematical Law



The Core Diagnostic Question:

Given a change in underlying data representation, when does a function's behavior transport automatically, and when does it require an explicit preservation hypothesis?



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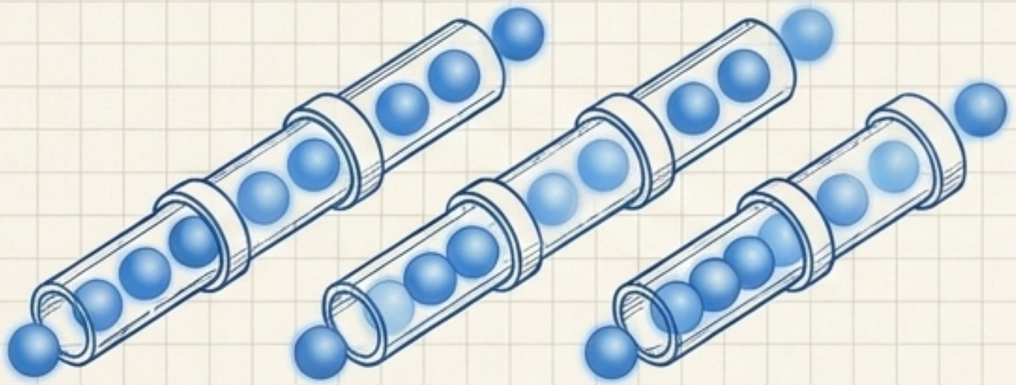
The Parametricity Spectrum

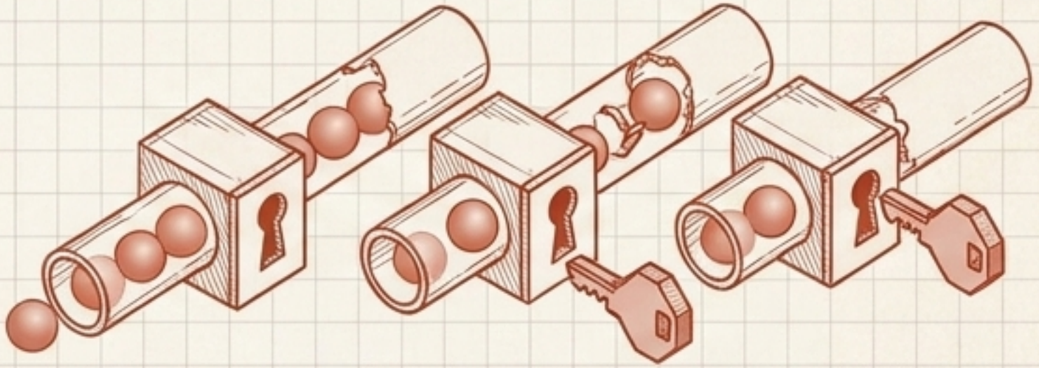
Architectural

Architectural

Free / Unconditional

Hypothesis-Bound





head tail fst

Yields relational consequences automatically.
Indifferent to surrounding representation.
No preservation assumptions required.

sort filter fold

Explicitly consumes additional structure (an ordering, a predicate). Fails unless a specific preservation obligation is met.

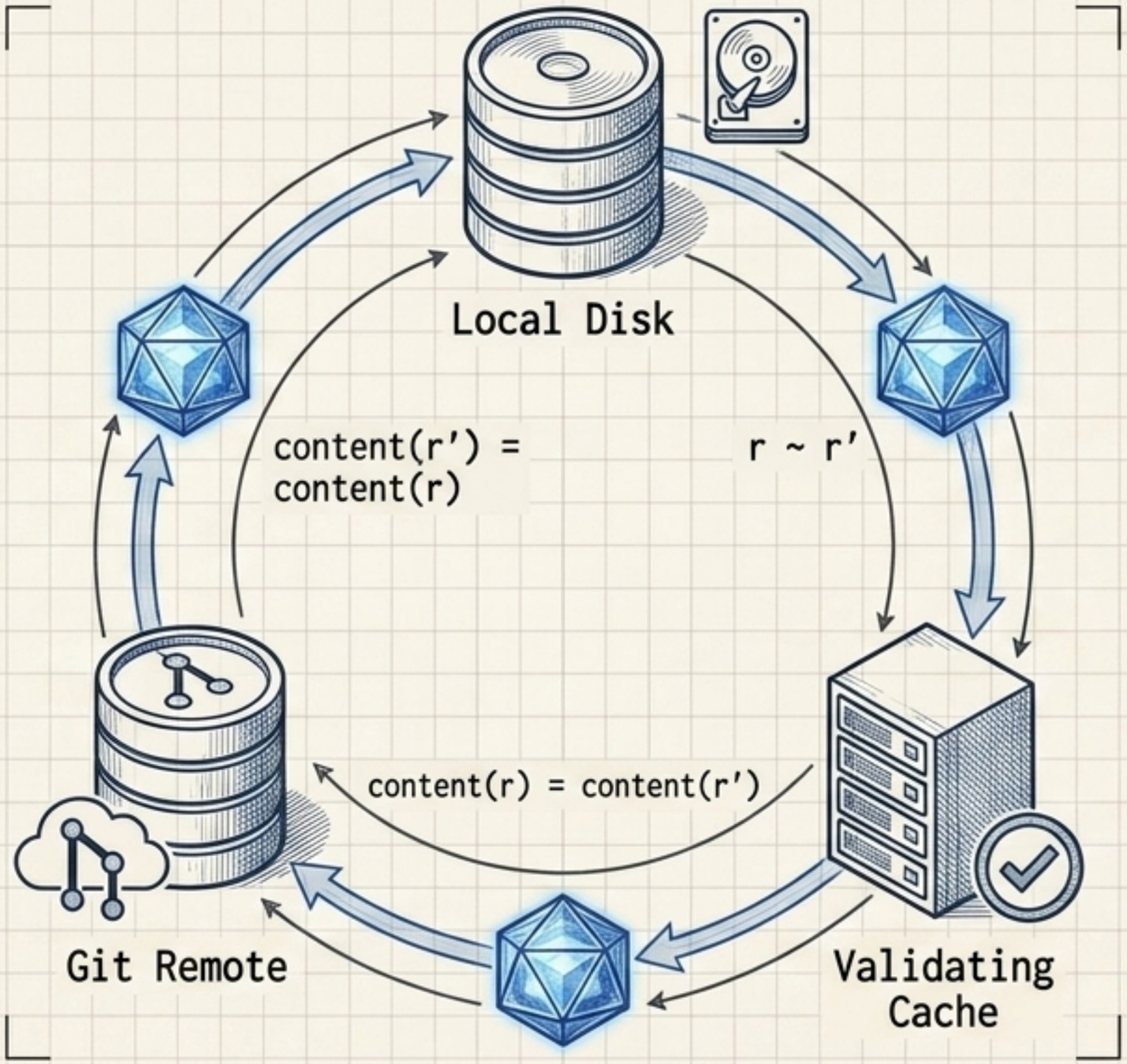
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Read and Save as Free Theorems

Architectural

Type Analogue:

RESOLVE and SAVE behave mathematically like head and fst.



Architectural

The Result:

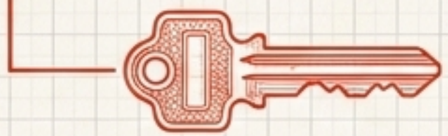
Their specifications are silent on where content physically lives. As long as content identity is maintained, storage representation can vary freely. Zero residency required.

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Auto-Triggering as the “Sort” Case

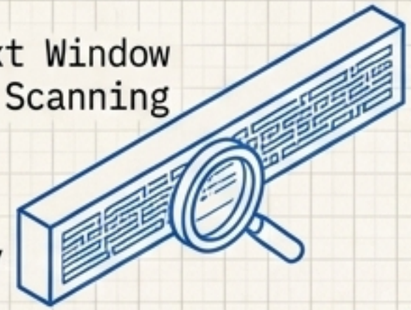
The Mathematical Requirement

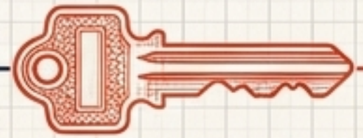
$$a* \circ sortA(<) = sortA'(<') \circ a*$$



Structural Key: An Order

The Architectural Requirement





Structural Key: Resident Prompt

TRIGGER

Auto-triggering is NOT free. Just as sort is mathematically defined by an order, TRIGGER in this architecture is defined by prompt residency. The token cost purchases this necessary residency hypothesis.

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The Interface Diagnostic Matrix

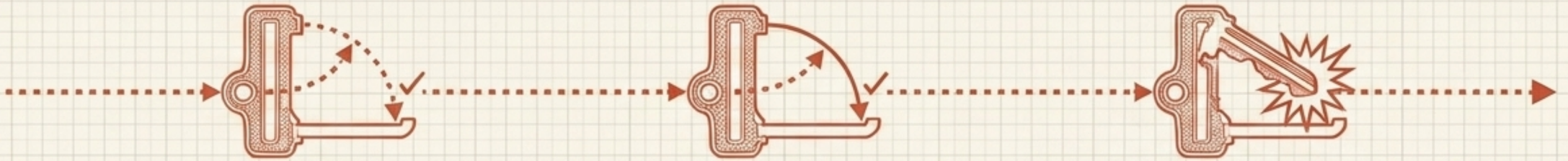
Operation	Type-Theory Analogue	Semantic Invariant (P_{sem})	Representational Obligation (P_{rep})	Standing Token Cost
RESOLVE	head	Content Identity	∅	Zero
SAVE	fst	Persistence	∅	Zero
TRIGGER (prompt)	sort	Unprompted Activation	{ Residency }	50-100 Tokens

Free-theorem reasoning allows us to mathematically isolate which operations owe an architectural debt.

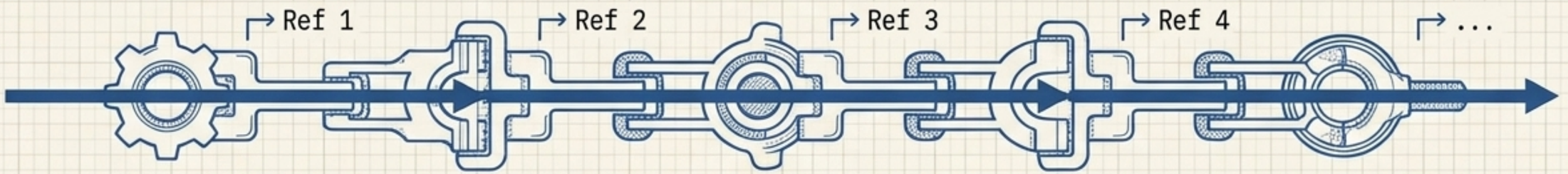
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Composition: Determinism vs. The Trigger Lottery

Installed Descriptions / Probabilistic



Explicit References / Deterministic



Uniform propagation via fold

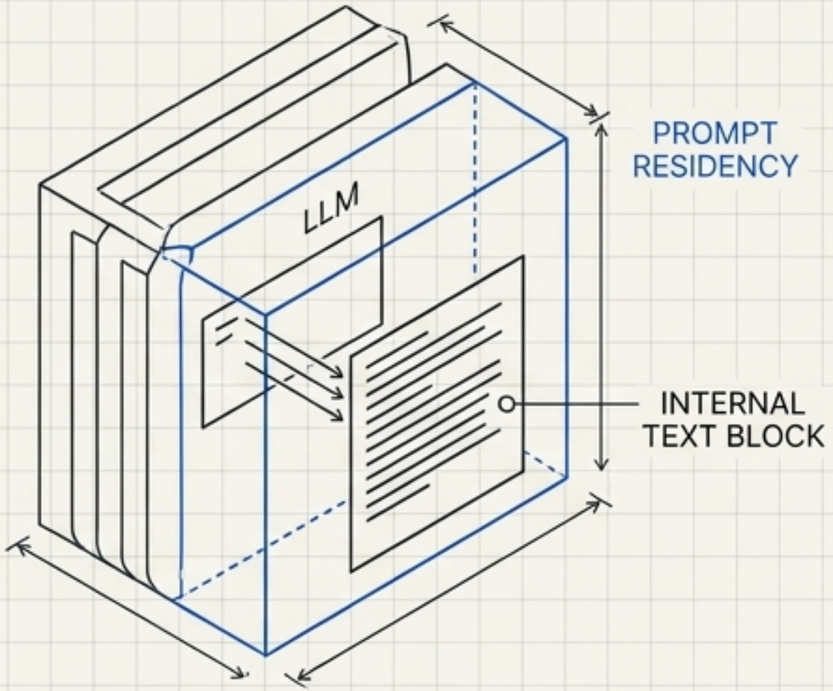
Installed descriptions require independent, **probabilistic** matches, creating a **fragile trigger lottery**.

Explicit references name their own paths. Like a **fold** theorem, selection determinism composes cleanly without degrading.

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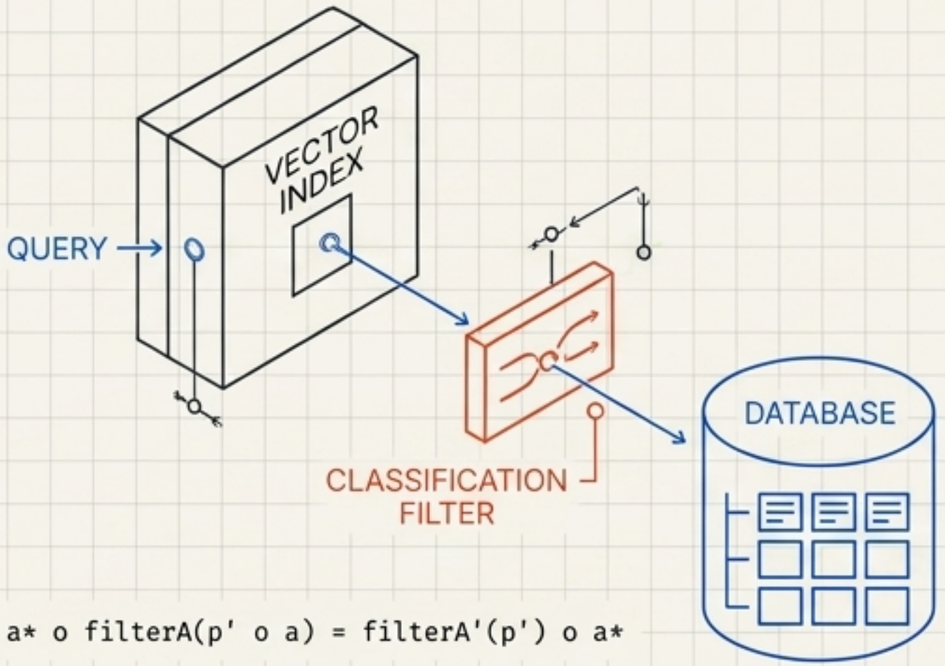
Testing the Framework: Retrieval-Mediated Triggers

Prompt-Mediated Triggering



Behaves like sort. The architectural obligation is prompt residency.

Retrieval-Mediated Triggering (RAG)



$a* \circ filterA(p' \circ a) = filterA'(p') \circ a*$

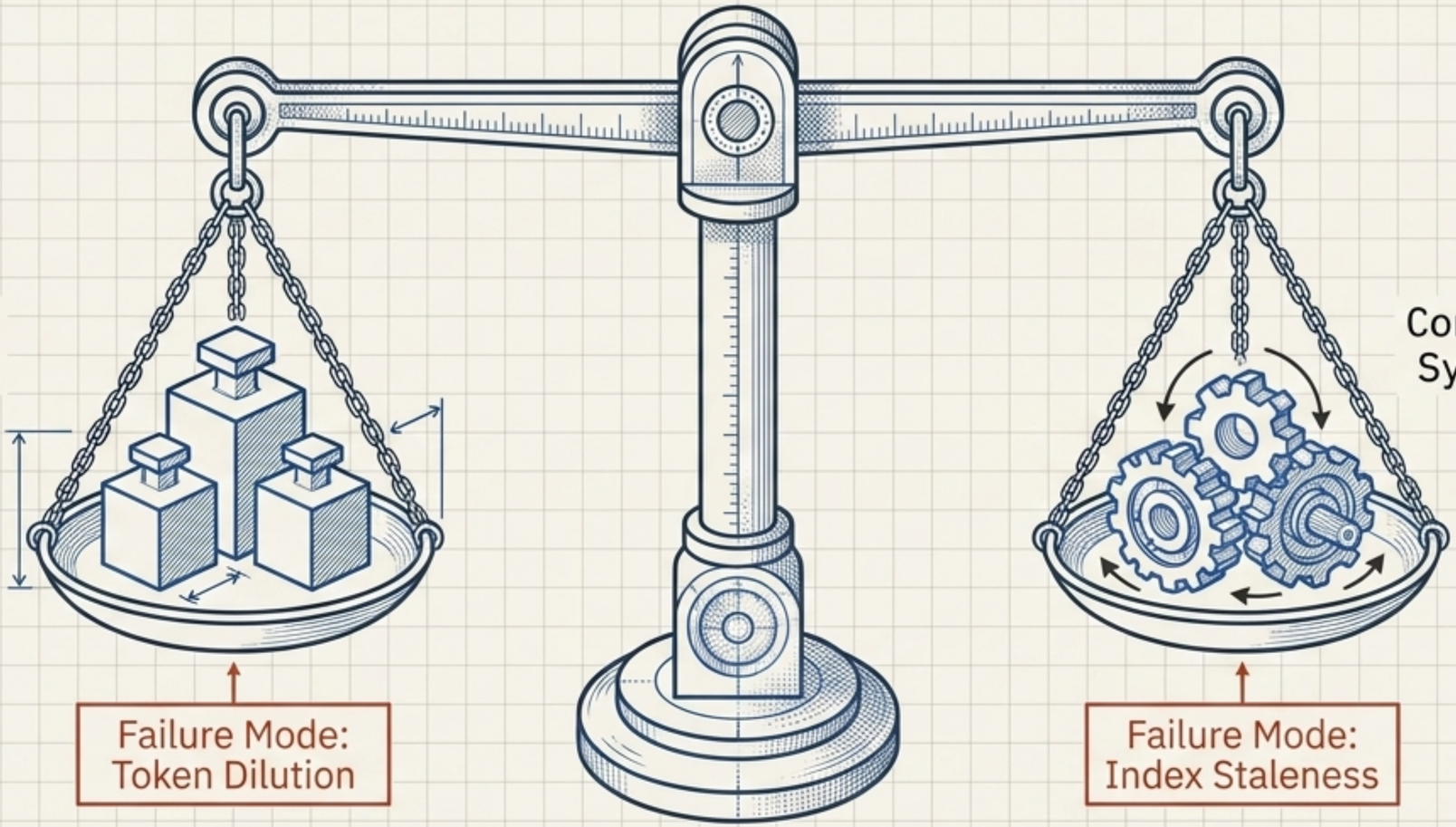
Behaves like filter. The architectural obligation changes to transporting the predicate.

Is prompt residency inherently required for triggering? No. When the architecture changes, the required mathematical hypothesis changes.

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The Conservation of Representational Obligation

Standing Token Cost



Failure Mode: Token Dilution

Continuous Index Synchronization

Failure Mode: Index Staleness

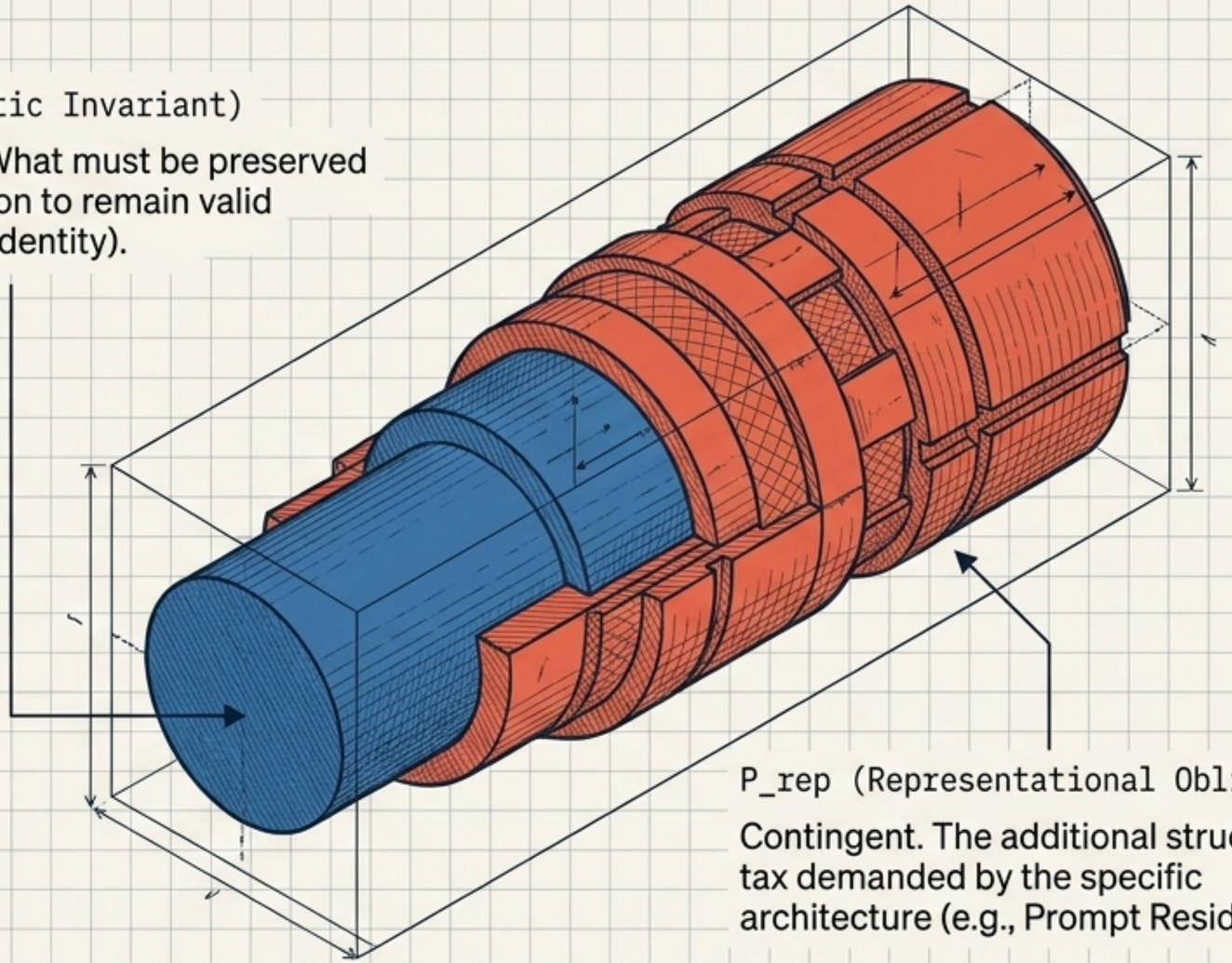
Architectural choices do not eliminate structural obligations; they merely relocate them.
A system must purchase either prompt occupancy or the transport of the retrieval predicate.

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Separating Invariants: P_sem vs P_rep

`P_sem (Semantic Invariant)`

Unavoidable. What must be preserved for the operation to remain valid (e.g., Content Identity).



`P_rep (Representational Obligation)`

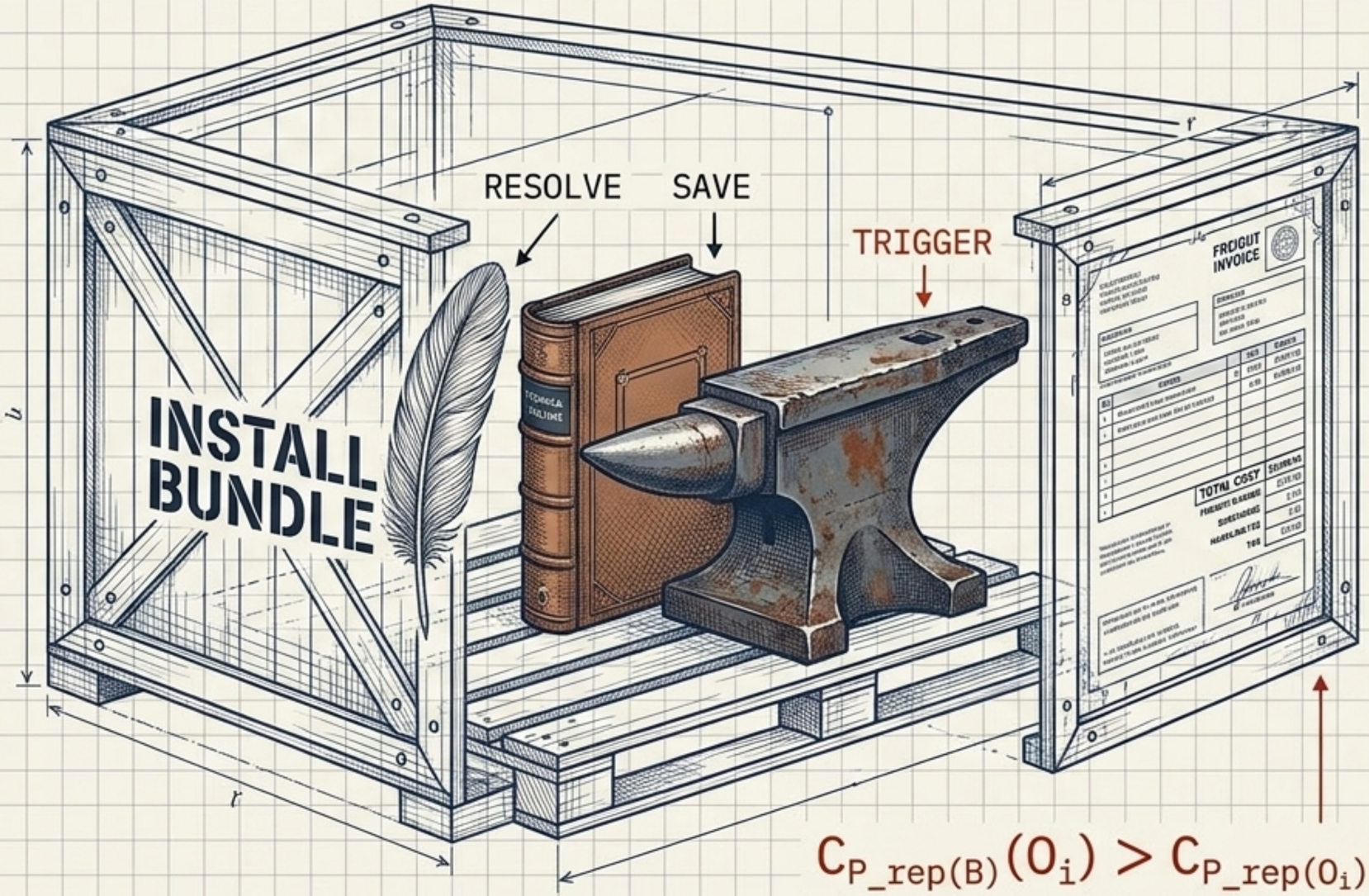
Contingent. The additional structural tax demanded by the specific architecture (e.g., Prompt Residency).

SUMMARY

- Every operation has a minimum semantic cost (`P_sem`).
- The goal of system design is to isolate and minimize the extra representational tax (`P_rep`).

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The Bundling Pathology



$$C_{P_rep(B)}(O_i) > C_{P_rep(O_i)}(O_i)$$

Commitment Leakage

The pathology occurs when a system prices a free operation at the rate of the bundle's most expensive operation.

You pay the architectural freight cost of the anvil even if you only need the feather.

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SYNTHESIS: The Law of Commitment Honesty

A system is commitment-honest when the cost of an operation tracks its own representational obligations, not the obligations of its bundled neighbors.

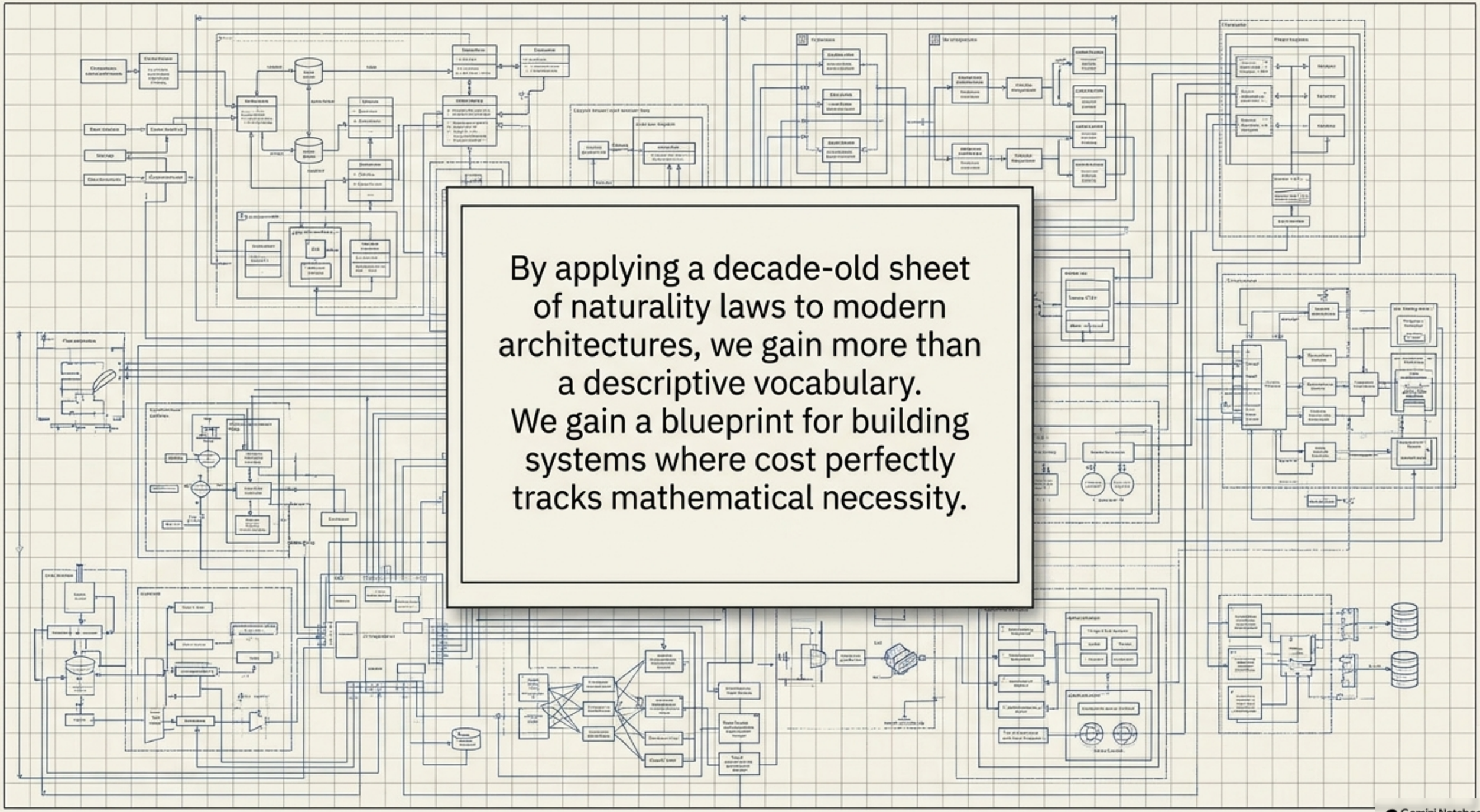
The Universal Diagnostic

This framework provides a rigorous vocabulary to locate bundling defects in any software system.

The Perfect Interface

A great interface charges exactly what the math demands—and absolutely nothing more.

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